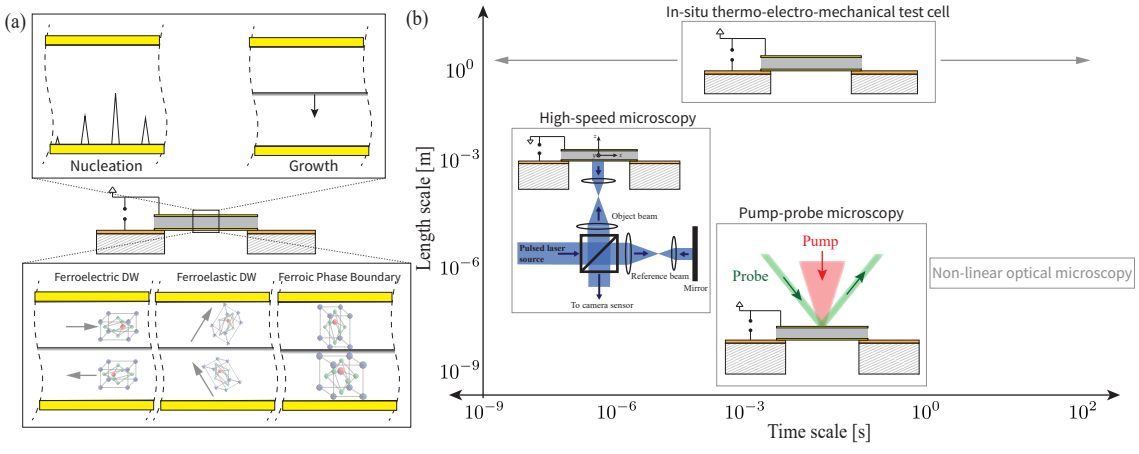

(a)
Nucleation
Growth
Ferroelectric DW
Ferroelastic DW
Ferroic Phase Boundary
(b)
In-situ thermo-electro-mechanical test cell
High-speed microscopy
Object beam
Pulsed laser source
Reference beam
Mirror
To camera sensor
Pump-probe microscopy
Pump
Probe
Non-linear optical microscopy
Length scale [m]
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-9}
10^{-6}
10^{-3}
10^{0}
10^{2}
Time scale [s]